



before. Its onboard quad-core ARM computer comes with Ubuntu 20.04 Aarch64 architecture OS and has built-in Neu-Viewer client software for scanning effect display and ROS driver software package supporting LiDAR. Moreover, the device comes with a Wi-Fi hotspot. A 17.7cm multitouch screen allows users to control the device without connecting their personal computers or other industrial personal computer. Applications include outdoor mapping, 3D imaging, and measurements for autonomous driving, smart city (V2X), drone, mapping, etc.

Neuvition
www.neuvition.com

Smart 3D surveillance system

To provide an enhanced and robust solution for high-security needs, the Leica BLK247—a 3D surveillance system—combines LiDAR, video camera, and thermal imaging technology for security that building owners, operators, and security teams can make use of for continuous monitoring of buildings and



spaces including hospitals, factories, manufacturing plants, and offices. Additionally, the Leica BLK247 provides reliable surveillance for private properties, allows creation of 3D zones by geofencing areas, and applies custom alarm rules. Assistive AI enables the device to make unaided real-time decisions by learning the space it is monitoring (known as a 'static scene') and through change detection technology recognises changes within that scene.

Leica Geosystems
www.leica-geosystems.com

5G and AI-enabled robotics platform

Designed specifically for robotics, the Robotics RB5 platform comprises an extensive set of hardware, software, and development tools. The Robotics RB5 platform brings together 5G and AI to



empower developers and manufacturers creating the next generation of high-compute, low-power robots and drones for the consumer, enterprise, defence, industrial, and professional service sectors. The platform's QRB5165 processor offers a powerful heterogeneous computing architecture coupled with an AI engine delivering 15 tera operations per second of AI performance for running complex AI and deep learning workloads. The processor also offers incredible machine learning inferencing at the edge. With support for 4G and 5G connectivity speeds via a companion module, the RB5 platform helps pave the way for the proliferation of 5G in robotics and intelligent systems.

Qualcomm Technologies
www.qualcomm.com

Contactless hi-tech access control

In order to secure a building or a workplace from harmful unseen organisms that can infect a person and put others at risk, the cloud-based access control system called SecurePass lets you



access and manage your property from a centralised location. Given the havoc caused by the pandemic, the installation of such a system is necessary. Simply place your smartphone in front of the facility, scan the QR code and enter your details. These details are then sent to the cloud through a centralised dashboard from anywhere in the world, where these are processed. On successful verification, access is granted. Once this happens, the system scans for body temperature and sends an alert or restricts access if high temperature is detected. SecurePass uses a combination of door sensors and people counter to detect and send alerts for all such scenarios.

Singularity Automation
www.thesecurepass.com

FPGA for satellite and space use

Delivering full radiation tolerance, ultra-high throughput and bandwidth performance for satellite and space applications, the KintexUltraScale XQRKU060 is a 20nm space-grade FPGA that provides



unlimited on-orbit reconfiguration and delivers a significant reduction in size, weight, and power. The XQRKU060 brings to space high-performance machine learning development tools that support industry-standard frameworks, including TensorFlow and PyTorch, enabling neural network inference acceleration for real-time on-board processing in space with a complete 'process and analyse' solution. The XQRKU060's dense, power-efficient compute with scalable precision and large on-chip memory provides 5.7 tera operations per second of peak INT8 performance, a nearly 25x increase compared to the prior generation, which is optimised for deep learning

Xilinx
www.xilinx.com